

## Smart Medication Dispenser

**Background:** Medication is easy to forget especially for someone with multiple medications at different times.

The goal is a smart pill dispenser that reminds someone when to take their medications and checks if a pill was taken. If a dose is missed the system records, it then it can notify the person/ caregiver. The goal is to make medication tracking automatic.

### Project needs

Design a smart medication dispenser that reminds user to take their medication

Add a weight sensing system that can detect when a pill was removed from its compartment

Develop a system/software that records medication schedules and whether a dose was taken or missed.

Create a backend so that medication information, schedules and history of doses can be recorded.

Design a front end to display dashboard of schedules and allow users to see whether a dose was taken or missed.

Add a notification feature so that users/caregivers are notified when a dose was taken or missed.

Connect dispenser, backend and frontend so dashboard information can be updated correctly and automatically.

### Project scope

Smart medication dispenser design and development

Medication schedule and reminder system

Weight sensing for detecting pills removed

Backend database for storing information and frontend for displaying it

System testing

### **Deliverables**

A functional prototype of the dispenser

A working backend and frontend system and dashboard for status, schedule and history

Documentation of design and system development

Test results and demo